

# Lecture 2

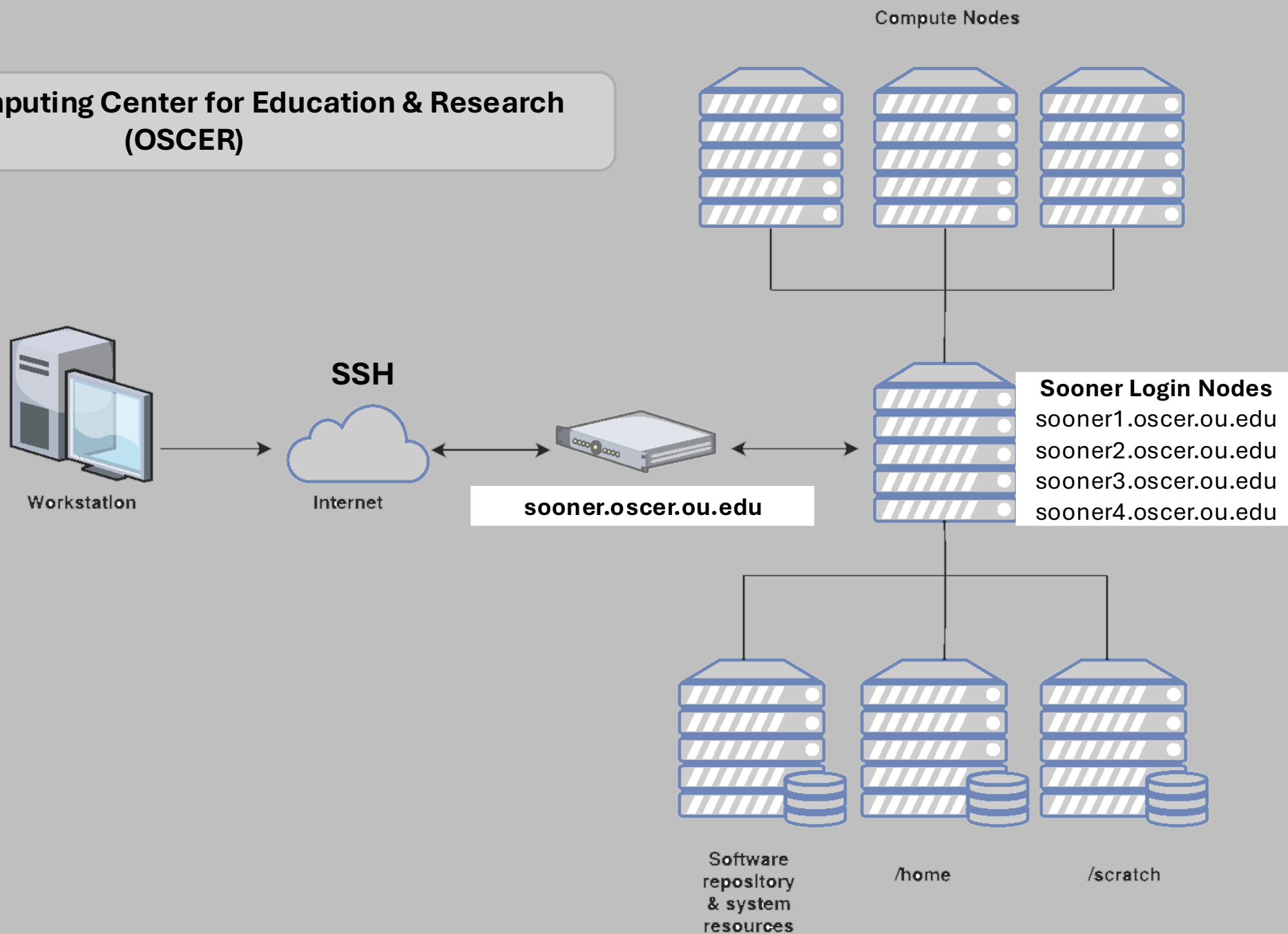
Eleana Cabello

# Course Website and Recorded Lectures

Website: <https://eleanacab.github.io/IntroToDataScience2026/>

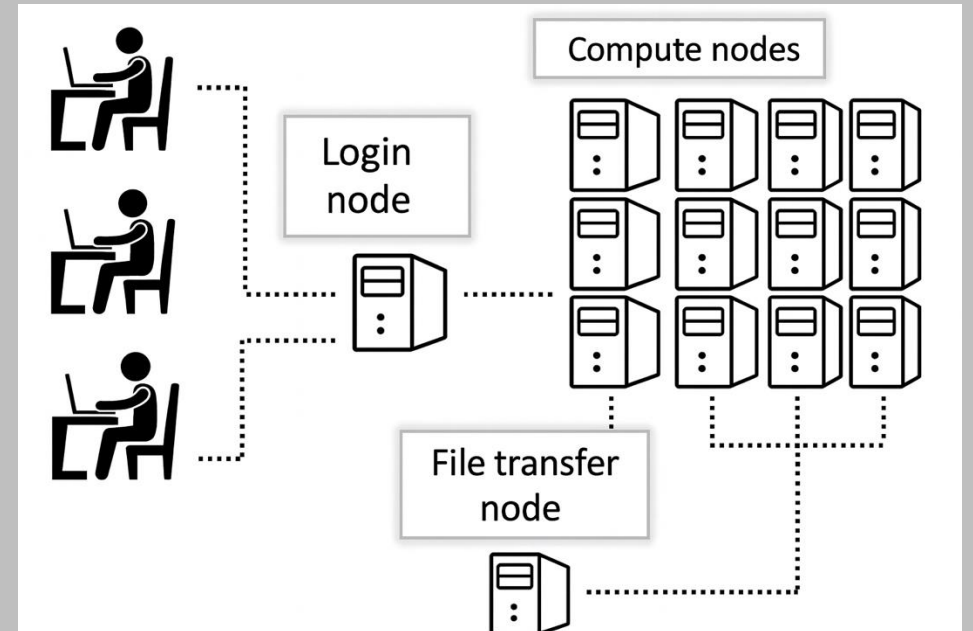
Recordings: [recorded\\_lectures](#)

# OU Supercomputing Center for Education & Research (OSCER)



# What is an HPC?

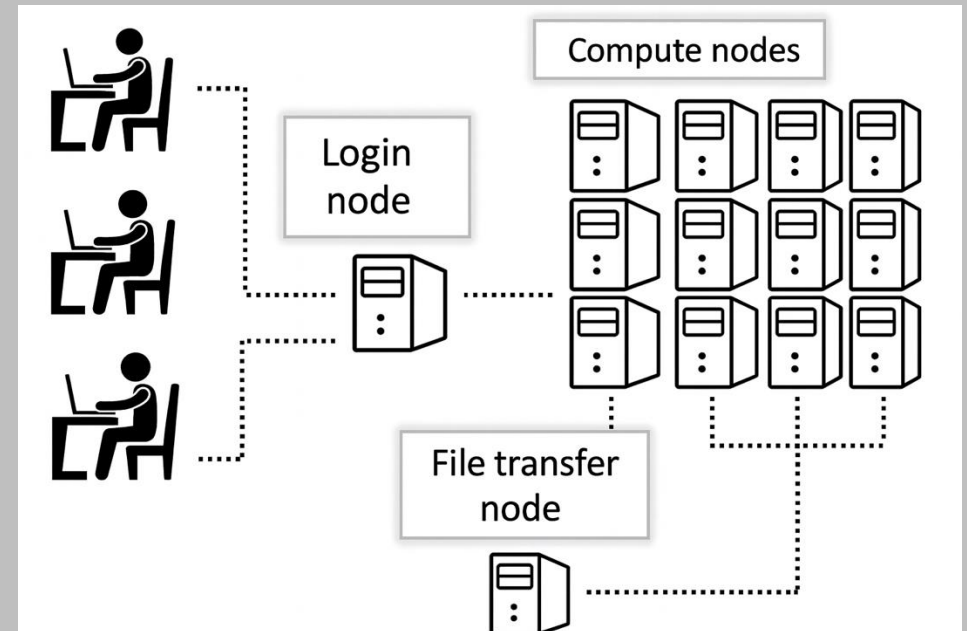
- HPC stands for **High Performance Computing**
- Supercomputer is another term for it
- It consists of many standard computers, known as **compute nodes**, connected through a network
- HPC systems operate as a shared resource for multiple users



# Nodes on HPC

**Login node:** Default access and entry point for all users

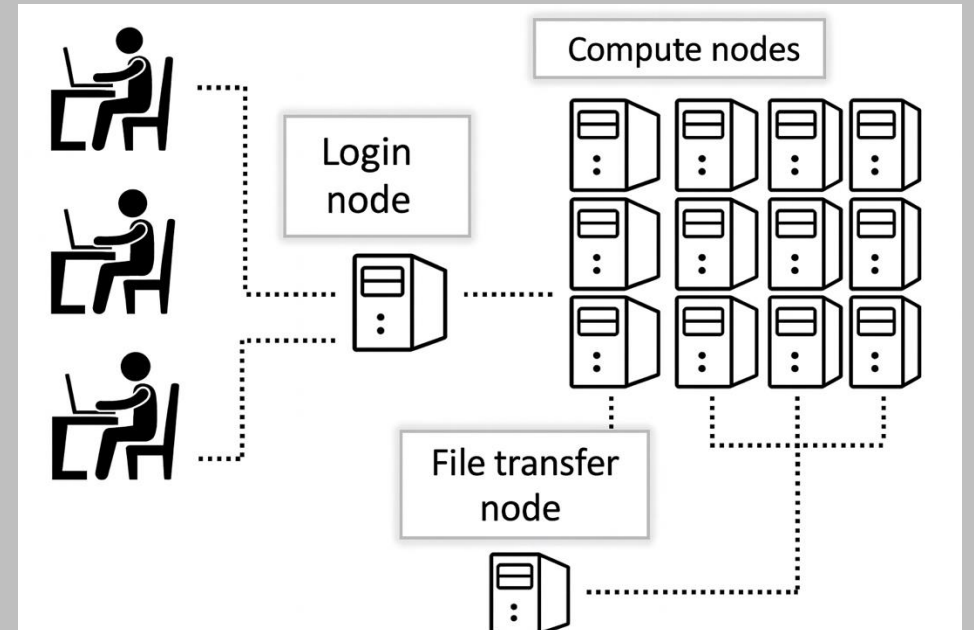
- Only used for file management and editing
- Not intended for heavy computations
- Intensive jobs on login nodes affect others and will be stopped by the HPC staff



# Nodes on HPC

**Compute nodes:** For running applications and executing computational tasks

- There are many compute nodes
- Each has large amounts of computational resources to run jobs
- Nodes are assigned to partitions



# Let's get on the same page

- Windows systems – Log onto OSCER using MobaXterm
- Mac systems - Find the Terminal application

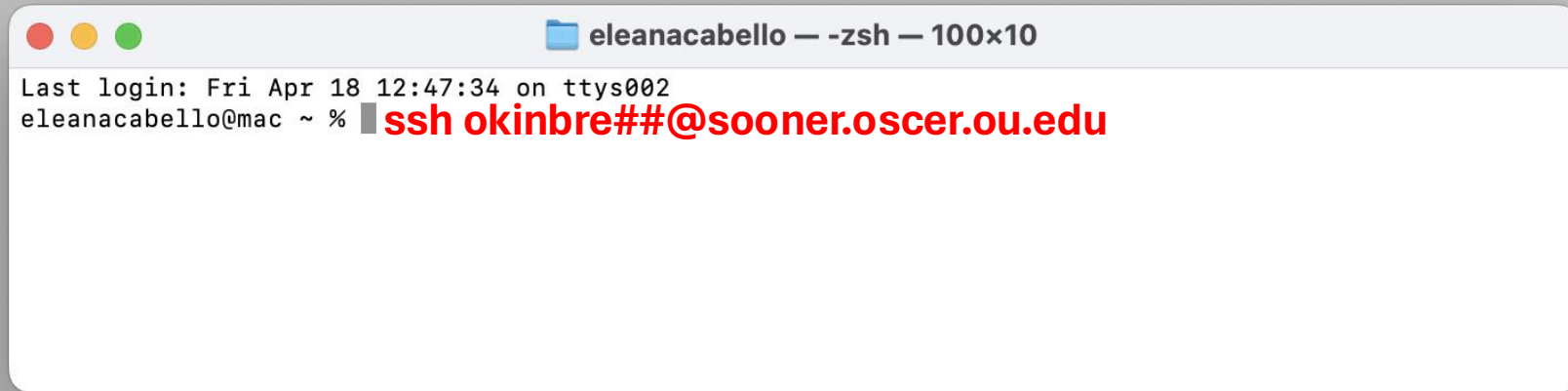


- **Terminal** - A text-based user interface that allows users to interact with a computer by typing commands
- **Directory** - A folder location in your computer
- **Working Directory** - The current directory a user is in
- **Path** - Directions to a folder or file



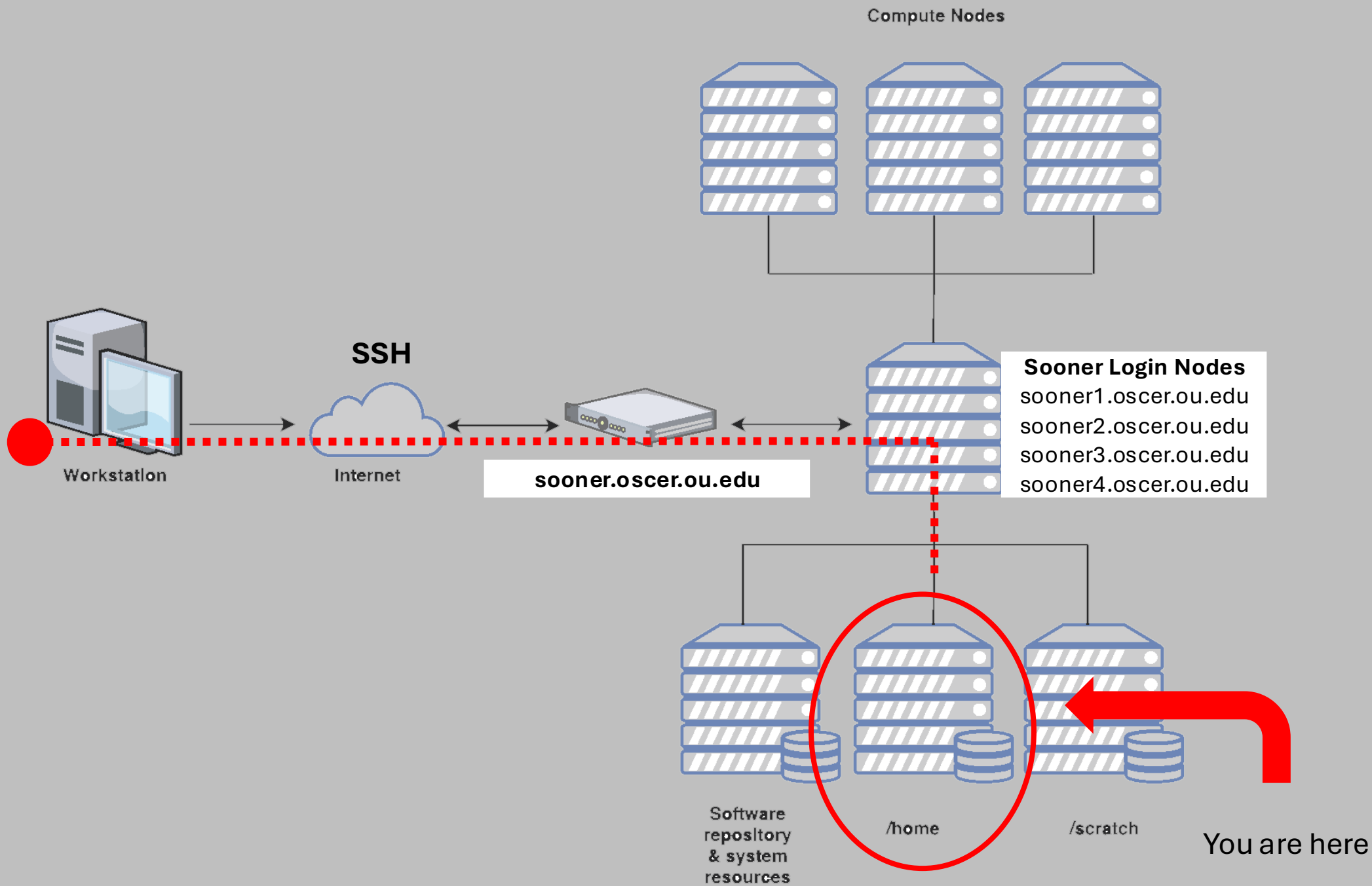
# Let's log into your guest accounts

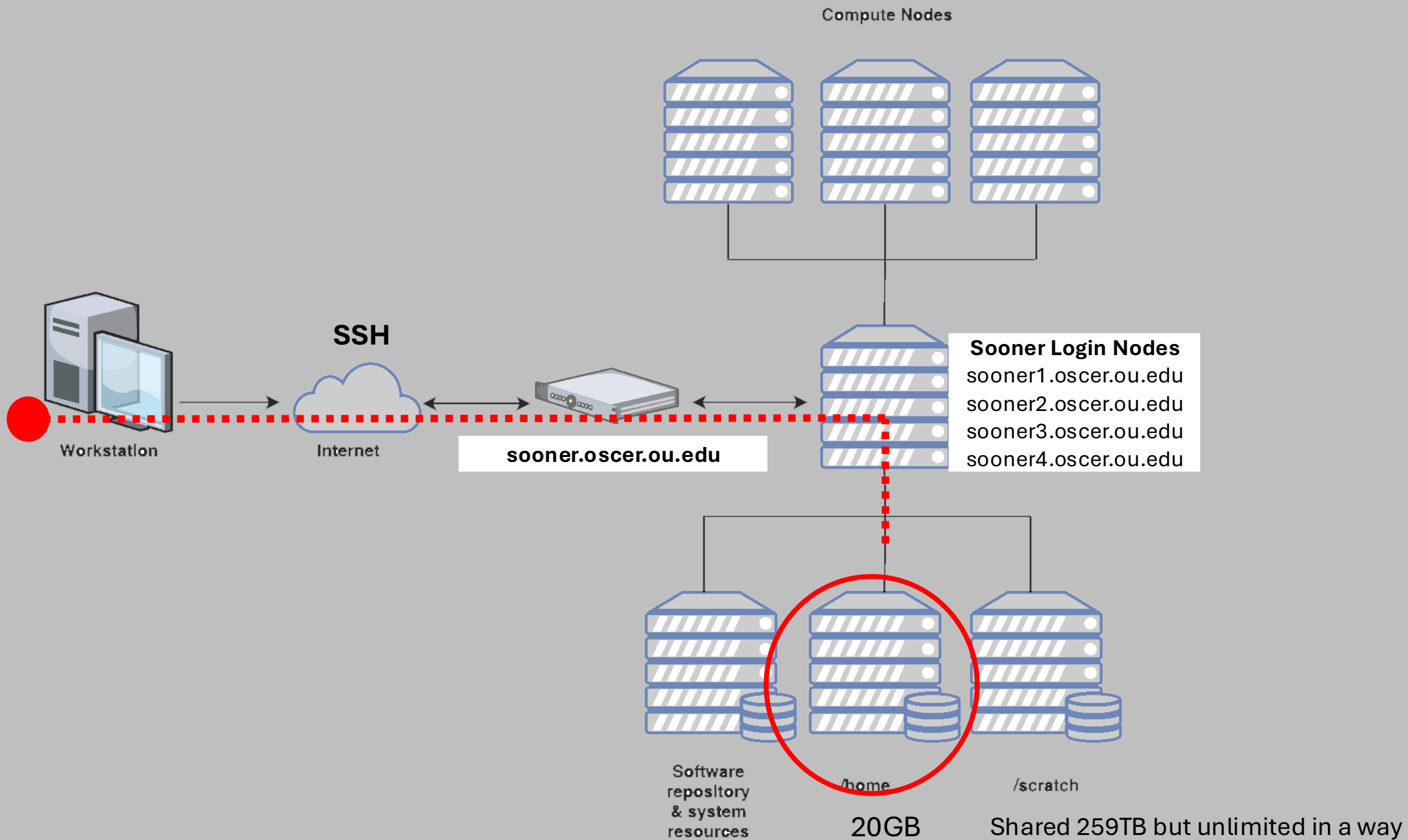
- Gather you log in information, username and password.

A screenshot of a macOS terminal window. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left, followed by a blue folder icon and the text "eleanacabello — -zsh — 100x10". The terminal content shows the output of a previous login: "Last login: Fri Apr 18 12:47:34 on ttys002" followed by the prompt "eleanacabello@mac ~ %". The user has entered the command "ssh okinbre##@sooner.oscer.ou.edu", which is displayed in red text. A grey cursor block is visible at the end of the command line.

```
eleanacabello — -zsh — 100x10
Last login: Fri Apr 18 12:47:34 on ttys002
eleanacabello@mac ~ % ssh okinbre##@sooner.oscer.ou.edu
```

- Type the following command to access you OSCER account: **ssh okinbre##@sooner.oscer.ou.edu**
- It will ask for your password, start typing it. It will not appear on your screen do not be alarmed this is normal. Just make sure to enter it correctly then hit enter.





# **BASH Programming**

From where you are type: **cp -r /scratch/elec/lecture\_2 /scratch/okinbre##**

Now type: **cd /scratch/okinbre##**

# Navigating Directories

|                     |                                                                                        |
|---------------------|----------------------------------------------------------------------------------------|
| <code>pwd</code>    | <code># Print working directory</code>                                                 |
| <code>ls</code>     | <code># List contents in a directory</code>                                            |
| <code>ls -lh</code> | <code># List contents in a directory in a list format and human read able sizes</code> |
| <code>tree</code>   | <code># List directory and file tree</code>                                            |
| <br>                |                                                                                        |
| <code>cd DIR</code> | <code># Change working directory to the entered directory</code>                       |
| <code>cd ..</code>  | <code># Change working directory to parent directory of current directory</code>       |
| <code>cd ~</code>   | <code># Change working directory to home</code>                                        |

# Creating, Moving, and Deleting Directories

```
mkdir DIR      # Create directory
```

```
rmdir DIR      # Remove Directory
```

```
rm -r DIR      # Remove Directory
```

```
mv ONE_DIR TWO_DIR  # Move directory to new directory
```



# Deleting, Copying, and Moving Files

```
rm FILE.txt           # Remove file  
cp FILE.txt FILE.txt  # Copy file to new directory  
mv FILE.txt FILE.txt  # Move file to new directory
```

**There is no back button once things are deleted or overwritten it is gone**

# Path and Command Line Structure

## Path Structure

```
cp -r /scratch/elec/lecture_2 /scratch/okinbre##
```



**/Path/to/file/or/directory**

Route to a file or directory/folder in the file system

## Command Structure

**[CMD] [-Flags/Options] [Input Files]**

**[CMD] --help**

**NOTE:** --help will always bring up the instructions and options for a command

**man [CMD]**

**NOTE:** man will bring up the manual for a command

# Command Line Text Editor

```
nano file_nano_test.txt      # Opens file in line for editing
```

Navigate with up, left, right, and down keys

Type as usual

Exit using CTRL+ X

```
Save modified buffer?
Y Yes
N No      ^C Cancel
```

The above message will appear because you made changes, click Y on your keyboard

```
File Name to Write: test.sh
^G Help      M-D DOS Format      M-A Append      M-B Backup File
^C Cancel    M-M Mac Format      M-P Prepend     ^T Browse
```

It will then ask about the filename you want to save it to, to accept click the ENTER key on your keyboard

```
#!/bin/bash
#SBATCH --partition=normal
#SBATCH --ntasks=1
#SBATCH --mem=1G
#SBATCH --job-name=jobname
#SBATCH --output=jobname_%J_%A_%a_stdout.txt
#SBATCH --error=jobname_%J_%A_%a_stderr.txt
#SBATCH --time=0:10:00
#SBATCH --mail-user=EMAIL@ou.edu
#SBATCH --mail-type=ALL
#SBATCH --chdir=/scratch/USER/lecture_2
```

## #SBATCH Directives

Where to send your job

Number of tasks

Memory needed

Name of job

Output file name

Error file name

Time limit

Email to send updates to, **PLEASE REMEMBER TO PUT YOURS**

All notifications

Working directory of the job

```
#=====
# BASH Strict mode (i.e. "fail fast" to reduce hard-to-find bugs)
set -e          # EXIT the script if any command returns non-zero exit status.
set -E          # Make ERR trapping work inside functions too.
set -u          # Variables must be pre-defined before using them.
set -o pipefail # If a pipe fails, returns the error code for the failed pipe
                # even if it isn't the last command in a series of pipes.
#=====
```

# Quick Exercise

# Batch Jobs

- User writes a script containing the instructions (code) that is sent to a compute node via the scheduler (SLURM)
- Resources (CPU, memory, time) must be requested
- Jobs run when resources are available
- No need to be physically present or monitor while your job is running
- Efficient use of shared resources
- Automates long or repetitive tasks

Change into your scratch: **cd /scratch/okinbre##**

Run this: **pwd**  **Make sure you are in /scratch/user/lecture\_2!**

Then type: **nano first\_exercise.sh**

```
#!/bin/bash
#SBATCH --partition=normal
#SBATCH --ntasks=1
#SBATCH --mem=1G
#SBATCH --job-name=jobname
#SBATCH --output=jobname_%J_%A_%a_stdout.txt
#SBATCH --error=jobname_%J_%A_%a_stderr.txt
#SBATCH --time=0:10:00
#SBATCH --mail-user=EMAIL@ou.edu
#SBATCH --mail-type=ALL
#SBATCH --chdir=/scratch/USER/lecture_2
```

 **CHANGE THIS TO YOUR EMAIL as in firstname-lastname@ou.edu**

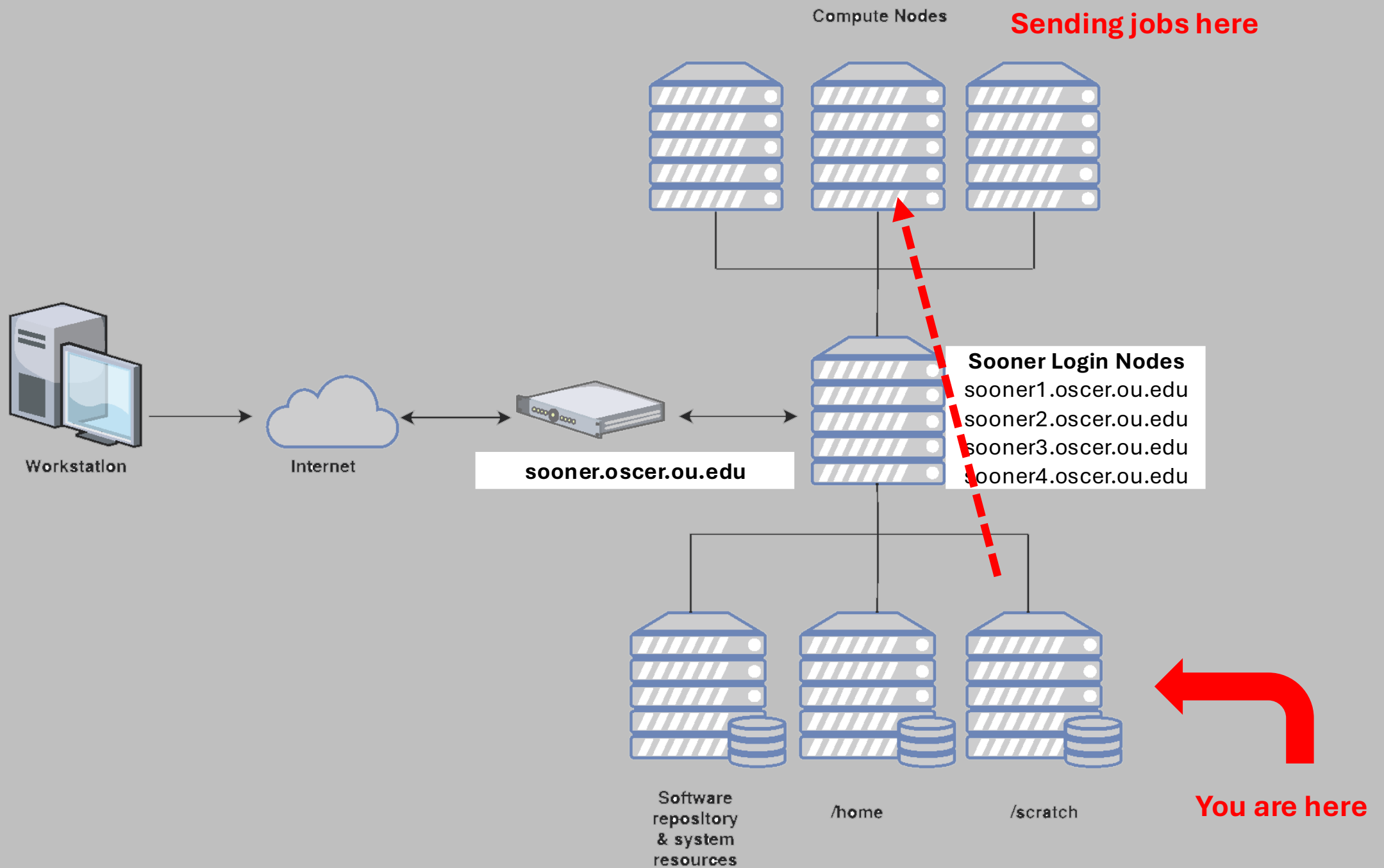
```
#=====
# BASH Strict mode (i.e. "fail fast" to reduce hard-to-find bugs)
set -e      # EXIT the script if any command returns non-zero exit status.
set -E      # Make ERR trapping work inside functions too.
set -u      # Variables must be pre-defined before using them.
set -o pipefail # If a pipe fails, returns the error code for the failed pipe
              # even if it isn't the last command in a series of pipes.
#=====
```

```
echo "Hello World!" > helloWorld.txt
```

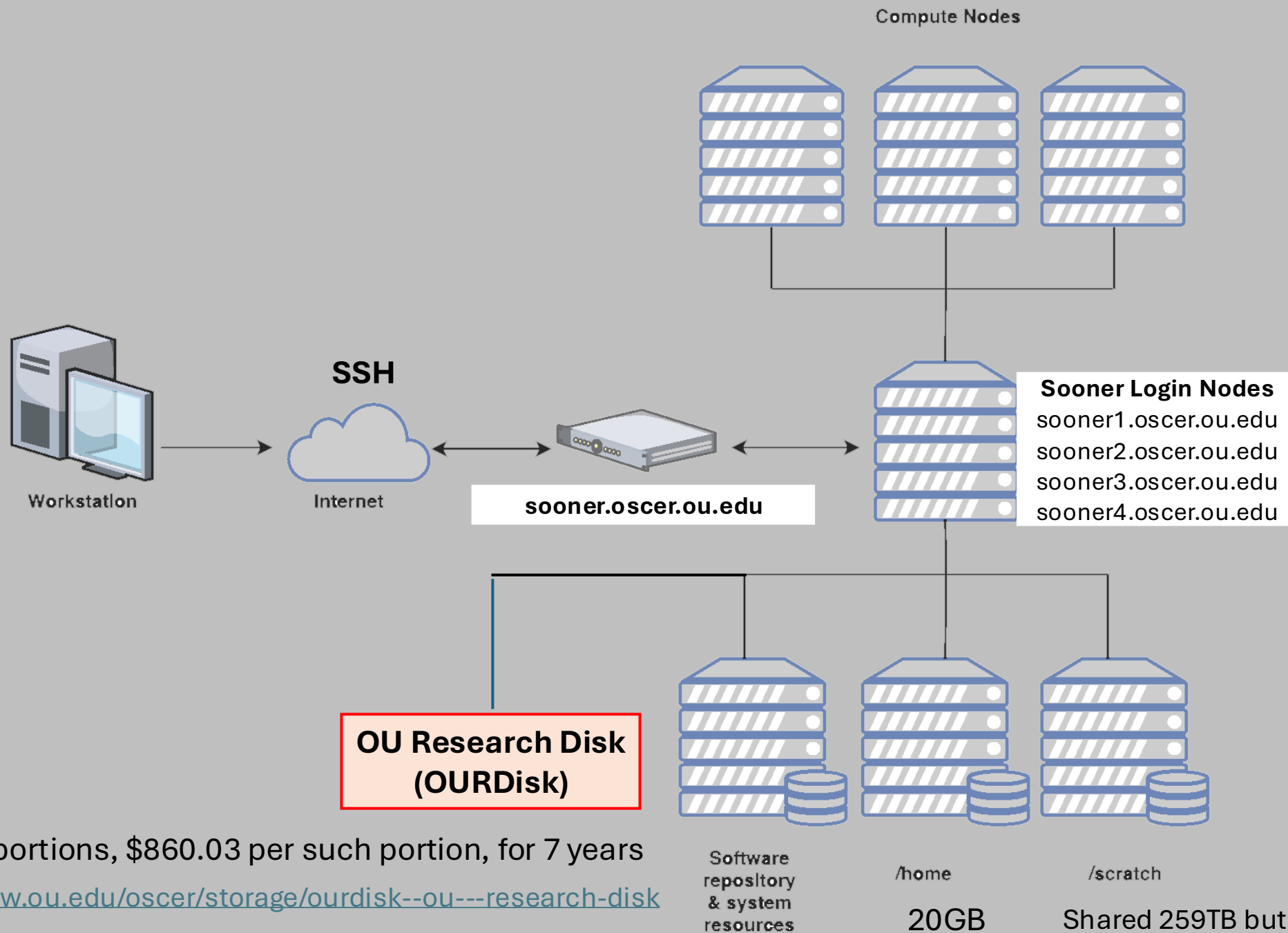
 **> redirects output into a file**



To submit job: **sbatch first\_exercise.sh**



# Useful Tips and Questions



~9.3 TB portions, \$860.03 per such portion, for 7 years

<https://www.ou.edu/oscer/storage/ourdisk--ou---research-disk>

# Before you leave today

- ❑ Have R and RStudio installed
- ❑ If possible, install needed packages:
  - [https://eleanacab.github.io/IntroToDataScience2026/resources/resources\\_R.html](https://eleanacab.github.io/IntroToDataScience2026/resources/resources_R.html)